Week 9

Spring 2027 · Section 029 · MW 9:00–10:20 AM · RRC Room 8210

Course Level Objectives:

- Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. (CLO 3)
- Explain the role of probability in statistics. (CLO 4)
- Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. (CLO 5)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Identify the parameter of interest in a research question. (CLO 8) **2.** Write null and alternative hypotheses in symbols and in words. (CLO 8) **3.** Classify a test as right-tailed, left-tailed, or two-tailed. (CLO 8) **4.** Explain the logic of a hypothesis test: assume H_0 is true and ask whether the sample data are surprising. (CLO 8)	e-Text: 4.1 Introducing Hypothesis Tests Interactive Lecture: 4.1 Introducing Hypothesis Tests Slides: 4.1 Introducing Hypothesis Tests	Participation: Writing Hypotheses – rock paper scissors practice (9 of 11 wins): define the parameter, state H_0 and H_a, choose the tail Your Turn: Write the Hypotheses (airline on-time rate, SAT prep, and more) Think about it: is 15 of 16 correct (dolphin study) convincing?	Online Homework: 4.1 Introducing Hypothesis Tests (due Sun Mar 28, 11:59 PM) Take-Home Quiz 6 (due Wed Mar 31, 11:59 PM) – covers 4.1–4.3 (all objectives) Test 2 (Fri Apr 2 – Sun Apr 4, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Apr 1) (all objectives)	WileyPLUS StatKey Blackboard Respondus
5. Describe how a randomization distribution is generated assuming H_0 is true. (CLO 5, CLO 8) **6.** Identify the center of a randomization distribution as the null value. (CLO 5) **7.** Compare sampling, bootstrap, and randomization distributions by their center and purpose. (CLO 5) **8.** Compute a p-value from a randomization distribution. (CLO 3, CLO 8) **9.** Determine the correct tail(s) for a p-value from the alternative hypothesis. (CLO 8) **10.** Interpret a p-value in context. (CLO 4, CLO 8)	e-Text: 4.2 Measuring Evidence with P-Values Interactive Lecture: 4.2 Measuring Evidence with P-Values Slides: 4.2 Measuring Evidence with P-Values StatKey – Randomization Hypothesis Tests	Participation: Randomization Distributions – StatKey: dolphin study (16 flips); count how many simulated statistics are as extreme as observed Your Turn: Tea and Immunity in StatKey (re-randomize tea vs. coffee, find the p-value)	Online Homework: 4.2 Measuring Evidence with P-Values (due Sun Mar 28, 11:59 PM) Take-Home Quiz 6 (due Wed Mar 31, 11:59 PM) – covers 4.1–4.3 (all objectives) Test 2 (Fri Apr 2 – Sun Apr 4, proctored) – covers P.1, 3.1–4.3 (all objectives) Test 2 Study Guide (all objectives) Practice Test 2 (optional; due Thu Apr 1) (all objectives)	WileyPLUS StatKey Blackboard Respondus